

DSA0602 Data Handling and Visualization

Unit II - Class Activity 1 Solutions & Visual Evaluation Report

Date: 20.07.2026 | Level: K5 Evaluation

Scenario 1: Hospital Patient Admission Analysis

Scenario Description: A hospital collected data on patient admissions across multiple departments (Cardiology, Neurology, Pediatrics, Orthopedics, Emergency) for the past year. Evaluate which visualization effectively communicates department workload, admission variability, and patient distribution.

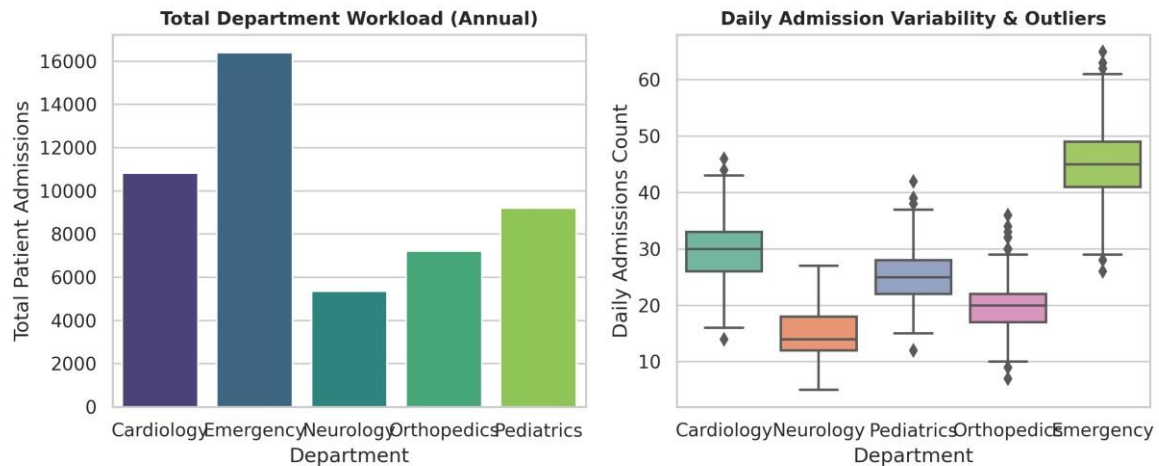
Suggested Libraries: Python: Pandas, Matplotlib, Seaborn, Plotly | R: ggplot2, dplyr **Python Implementation Code:**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Generate Synthetic Data
departments = ["Cardiology", "Neurology", "Pediatrics", "Orthopedics", "Emergency"]
dates = pd.date_range(start="2025-01-01", periods=365, freq="D")
data = []
for d in dates:
    for dept in departments:
        lam_val = 45 if dept == "Emergency" else (30 if dept == "Cardiology" else 20)
        data.append({"Date": d, "Department": dept, "Admissions": np.random.poisson(lam=lam_val)})
df = pd.DataFrame(data)

# Visualization
fig, axes = plt.subplots(1, 2, figsize=(10, 4.2))
total_adm = df.groupby("Department")["Admissions"].sum().reset_index()
sns.barplot(data=total_adm, x="Department", y="Admissions", ax=axes[0], palette="viridis")
sns.boxplot(data=df, x="Department", y="Admissions", ax=axes[1], palette="Set2")
plt.tight_layout()
plt.show()
```

Generated Visual Output Dashboard:



Scenario 2: E-Commerce Customer Spending Behaviour

Scenario Description: An online retailer has customer purchase amounts, order frequency, and product categories. Create multiple visualizations to identify high-value customers, spending variability, and purchasing trends. Recommend the best visualization for business executives.

Suggested Libraries: Python: Seaborn, Plotly Express / R:

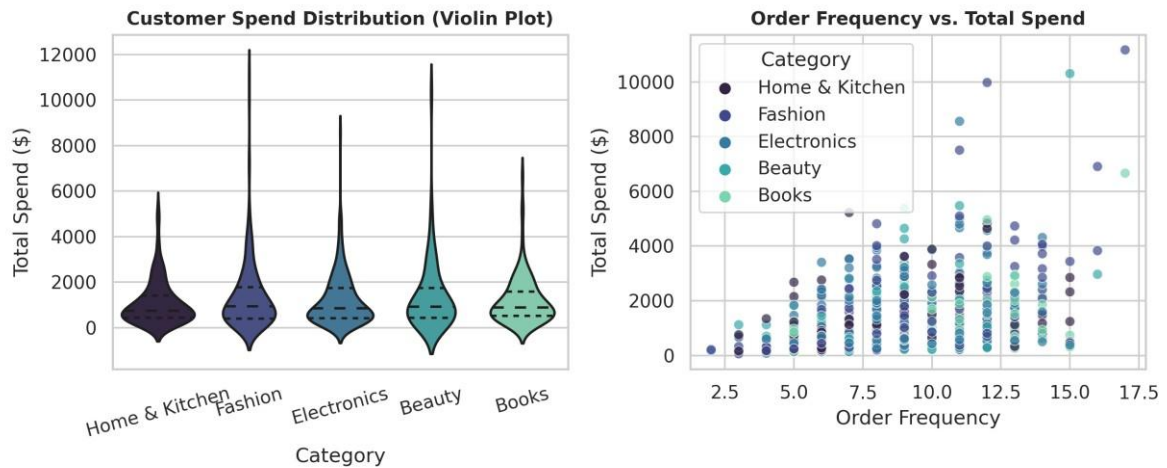
ggplot2 Python Implementation Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

categories = ["Electronics", "Fashion", "Home & Kitchen", "Books", "Beauty"]
df = pd.DataFrame({
    "Category": np.random.choice(categories, 800),
    "OrderFrequency": np.random.poisson(lam=8, size=800) + 1,
    "AvgPurchaseAmount": np.random.exponential(scale=120, size=800) + 20
})
df["TotalSpend"] = df["OrderFrequency"] * df["AvgPurchaseAmount"]

fig, axes = plt.subplots(1, 2, figsize=(10, 4.2))
sns.violinplot(data=df, x="Category", y="TotalSpend", palette="mako", inner="quartile", ax=axes[0])
sns.scatterplot(data=df, x="OrderFrequency", y="TotalSpend", hue="Category", palette="mako", ax=axes[1])
plt.tight_layout()
plt.show()
```

Generated Visual Output Dashboard:



Scenario 3: University Examination Performance Dashboard

Scenario Description: A university wants to compare student marks across departments and identify score distributions, outliers, and top-performing classes. Evaluate chart options for academic decision-making. **Suggested Libraries:** Python: Pandas, Seaborn, Plotly / R: ggplot2, plotly

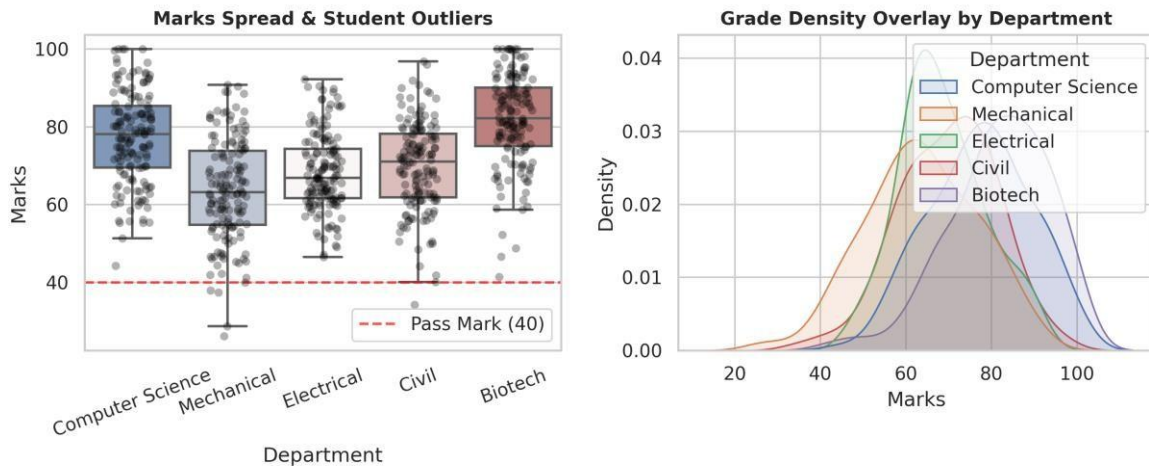
Python Implementation Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

depts = ["Computer Science", "Mechanical", "Electrical", "Civil",
"Biotech"]
records = []
for d in depts:
    scores = np.clip(np.random.normal(70, 12, 150), 0, 100)
    for s in scores:
        records.append({"Department": d, "Marks": s})

df = pd.DataFrame(records)
fig, axes = plt.subplots(1, 2, figsize=(10, 4.2))
sns.boxplot(data=df, x="Department", y="Marks", ax=axes[0], palette="vlag", fliersize=0)
sns.stripplot(data=df, x="Department", y="Marks", ax=axes[0], color="black", alpha=0.3)
axes[0].axhline(40, color="red", linestyle="--")
sns.kdeplot(data=df, x="Marks", hue="Department", fill=True, alpha=0.15, ax=axes[1])
plt.tight_layout()
plt.show()
```

Generated Visual Output Dashboard:



Scenario 4: Climate and Rainfall Distribution Study

Scenario Description: Meteorological data contains monthly rainfall, temperature, and humidity for multiple cities. Evaluate which visualization best represents seasonal rainfall variation and climate distribution for researchers.

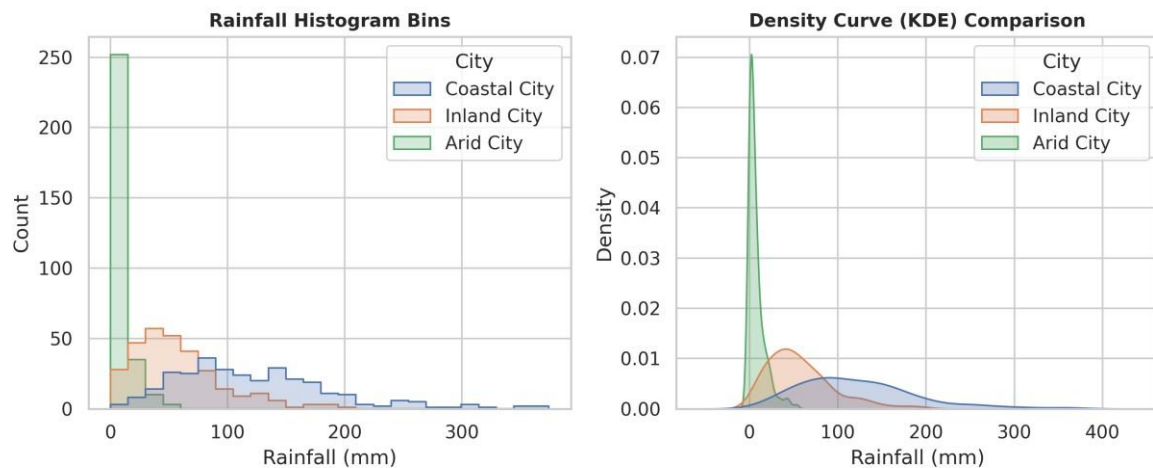
Suggested Libraries: Python: Matplotlib, Seaborn / R:

ggplot2 Python Implementation Code:

```
import pandas as pd
import numpy as np
import
matplotlib.pyplot as
plt import seaborn
as sns
```

```
cities = ["Coastal City", "Inland City", "Arid
City"] records = [] for city in cities:
    shape_val = 3.0 if "Coastal" in city else (0.5 if "Arid" in city else
2.0)    rf = np.random.gamma(shape=shape_val, scale=30, size=300)    for
val in rf:
        records.append({"City": city, "Rainfall_mm": val})
df = pd.DataFrame(records)
fig, axes = plt.subplots(1, 2, figsize=(10, 4.2)) sns.histplot(data=df,
x="Rainfall_mm", hue="City", element="step", ax=axes[0]) sns.kdeplot(data=df,
x="Rainfall_mm", hue="City", fill=True, alpha=0.3, ax=axes[1]) plt.tight_layout()
plt.show()
```

Generated Visual Output Dashboard:



Scenario 5: Bank Loan Risk Assessment

Scenario Description: A financial institution has loan amounts, repayment history, income levels, and credit scores. Evaluate visualizations showing loan distributions and customer segments supporting approval decisions. **Suggested Libraries:** Python: Seaborn, Plotly, Pandas | R: ggplot2

Python Implementation Code:

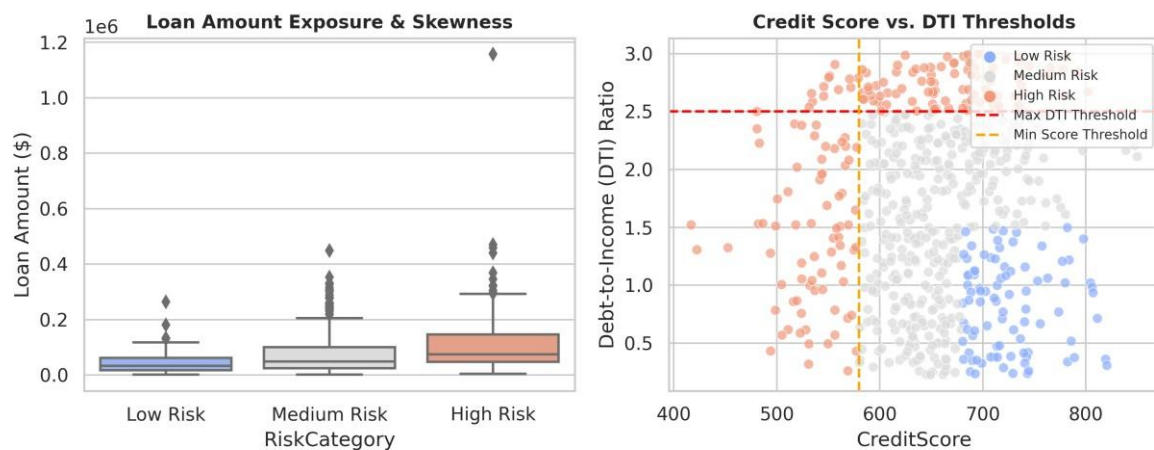
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

income = np.random.lognormal(mean=10.5, sigma=0.75, size=600)
loan_amount = income * np.random.uniform(0.2, 3.0, size=600)
credit_score = np.random.normal(650, 70, size=600).clip(300, 850)
dti = loan_amount / income

risk = ["High Risk" if s < 580 or r > 2.5 else ("Medium Risk" if s < 680 or r > 1.5 else "Low Risk") for s, r in zip(credit_score, dti)]
df = pd.DataFrame({"LoanAmount": loan_amount, "CreditScore": credit_score, "DTI": dti, "RiskCategory": risk})

fig, axes = plt.subplots(1, 2, figsize=(10, 4.2))
sns.boxplot(data=df, x="RiskCategory", y="LoanAmount", palette="coolwarm", ax=axes[0])
sns.scatterplot(data=df, x="CreditScore", y="DTI", hue="RiskCategory", palette="coolwarm", ax=axes[1])
axes[1].axhline(2.5, color="red", linestyle="--")
axes[1].axvline(580, color="orange", linestyle="--")
plt.tight_layout()
plt.show()
```

Generated Visual Output Dashboard:



R Programs with Output Screenshots (Q1-Q5)

Q1 Hospital Patient Admission Analysis

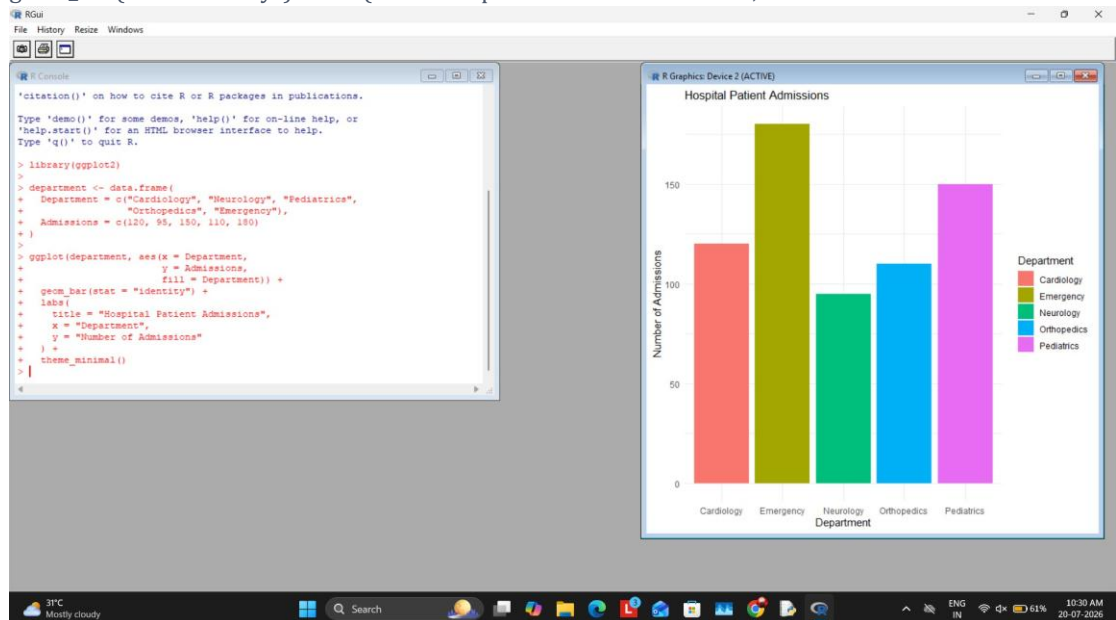
```
library(ggplot2)
```

```
department <- data.frame(
```

```
  Department=c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency"),
```

```
  Admissions=c(120,95,150,110,180) )
```

```
ggplot(department,aes(x=Department,y=Admissions,fill=Department))+  
  geom_bar(stat="identity")+ labs(title="Hospital Patient Admissions",
```



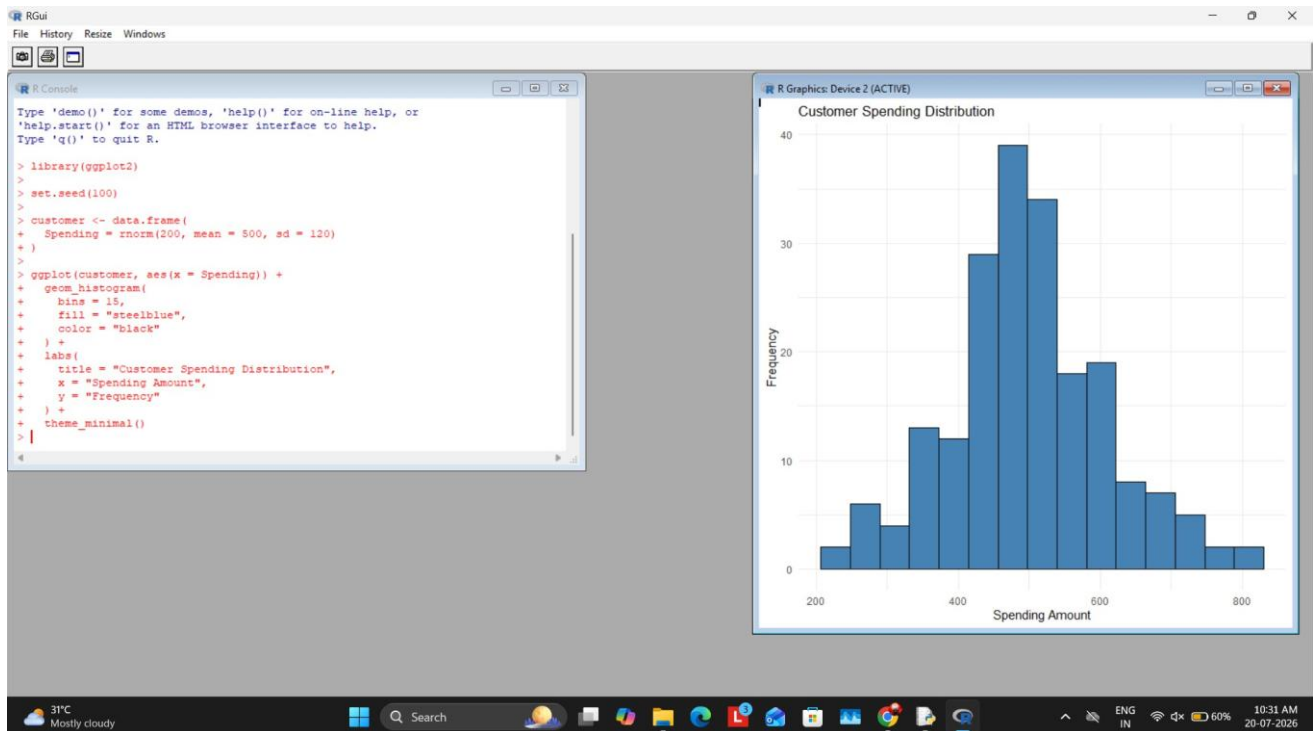
```
x="Department", y="Number of Admissions")+ theme_minimal()
```

Q2 E-Commerce Customer Spending Behaviour

```
library(ggplot2) set.seed(100)
```

```
customer<-data.frame(  
  Spending=rnorm(200,mean=500,sd=120) )
```

```
ggplot(customer,aes(x=Spending))+  
  geom_histogram(bins=15,fill="steelblue",color="black")+  
  labs(title="Customer Spending Distribution", x="Spending  
Amount", y="Frequency")+ theme_minimal()
```

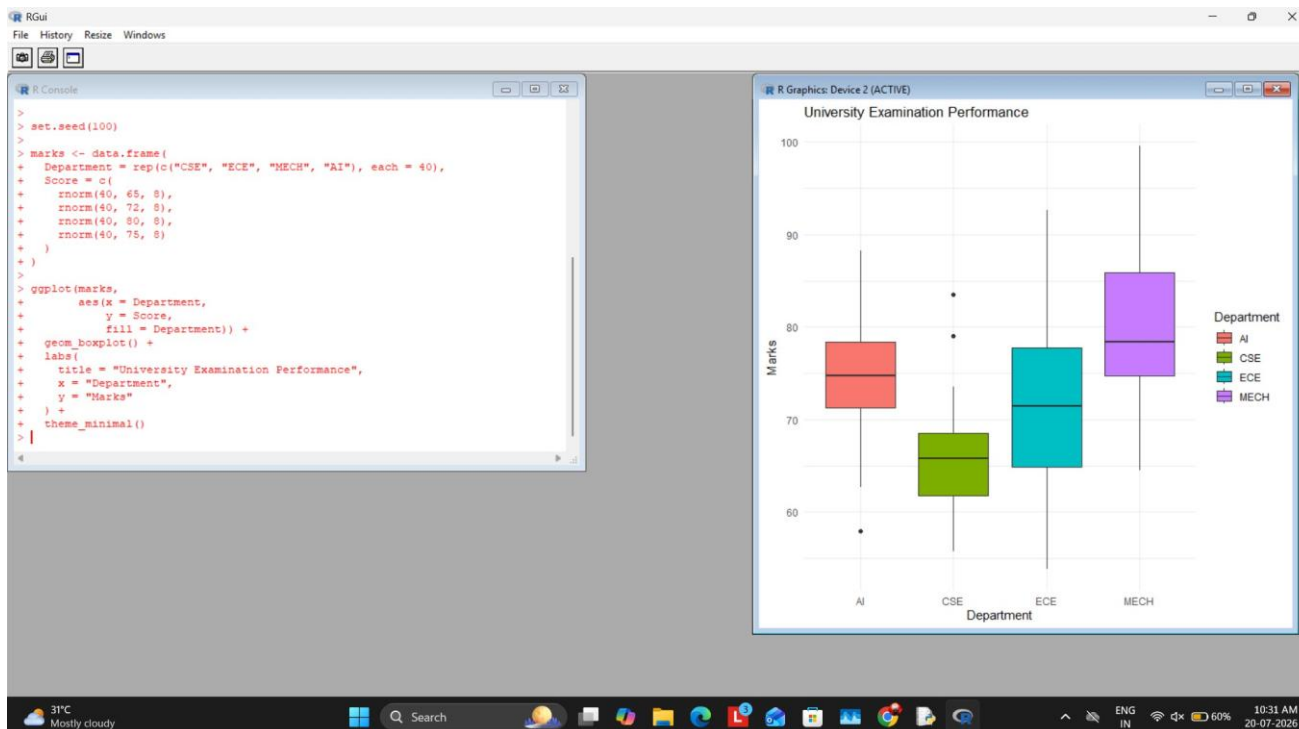


Q3 University Examination Performance Dashboard

```
library(ggplot2) set.seed(100)
```

```
marks<-data.frame(  
  Department=rep(c("CSE","ECE","MECH","AI"),each=40),  
  Score=c(rnorm(40,65,8), rnorm(40,72,8),  
          rnorm(40,80,8), rnorm(40,75,8)) )
```

```
ggplot(marks,aes(x=Department,y=Score,fill=Department))+ geom_boxplot()+  
  labs(title="University Examination Performance",  
        x="Department", y="Marks")+ theme_minimal()
```



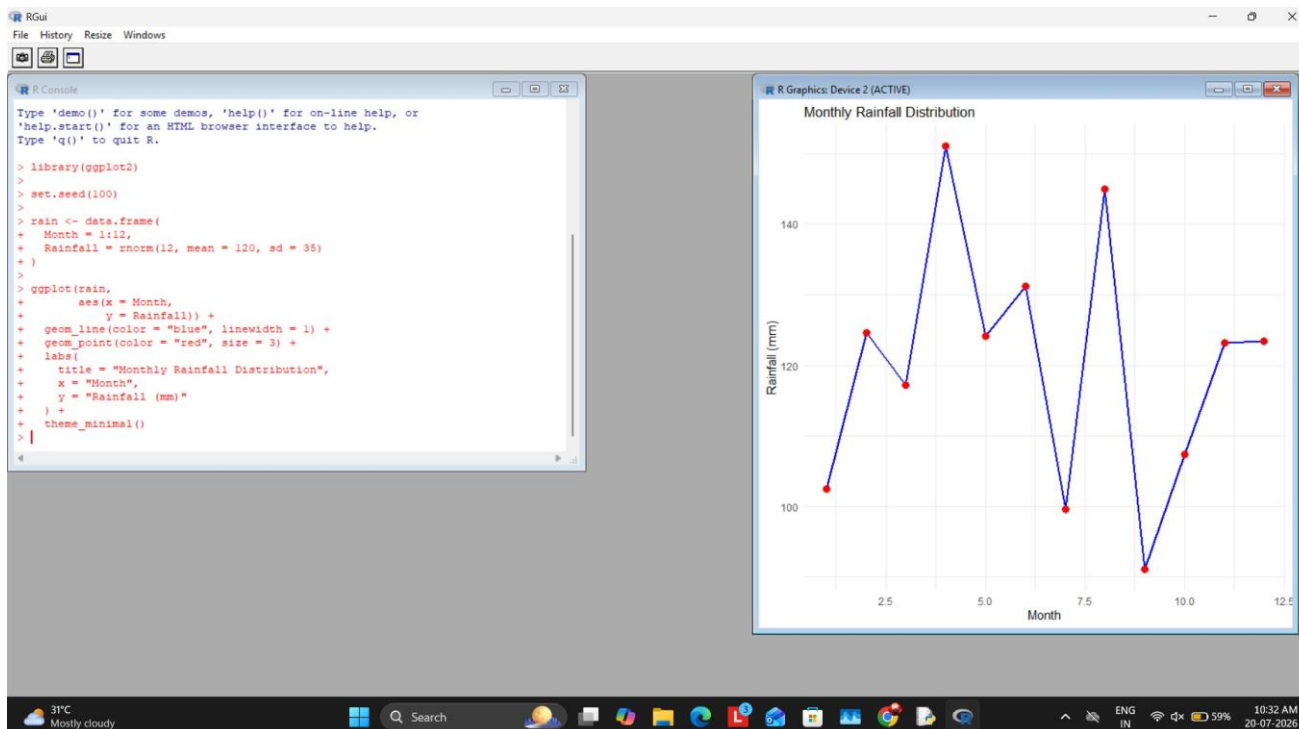
Q4 Climate and Rainfall Distribution Study

```
library(ggplot2) set.seed(100)
```

```
rain<-data.frame( Month=1:12,
```

```
Rainfall=rnorm(12,mean=120,sd=35) )
```

```
ggplot(rain,aes(x=Month,y=Rainfall))+
  geom_line(color="blue",linewidth=1)+
  geom_point(color="red",size=3)+
  labs(title="Monthly Rainfall Distribution",
       x="Month", y="Rainfall (mm)")+ theme_minimal()
```

Q5 Bank Loan Risk Assessment

```
library(ggplot2) set.seed(100)
```

```
loan<-data.frame(
  LoanAmount=rgamma(200,shape=2,scale=5000) )
```

```
ggplot(loan,aes(x=LoanAmount))+
  geom_histogram(bins=15,fill="tomato",color="black")+
  labs(title="Loan Amount Distribution", x="Loan Amount",
  y="Frequency")+ theme_minimal()
```

